

1(a)(i)	any one from: <u>no chloroplasts</u> (present) ; presence of (many) small vacuoles / no large vacuole / no central vacuole ; central nucleus or nucleus not located at the, edge / periphery ;	1																							
1(a)(ii)	one mark per row ;;; If 0 marks, allow 1 mark if name of cell structure column completed correctly <table border="1"> <thead> <tr> <th>function</th><th>name of cell structure</th><th>letter on Fig.1.1</th></tr> </thead> <tbody> <tr> <td rowspan="2">gas exchange</td><td>cell <u>surface</u> membrane I phospholipid bilayer</td><td>A</td></tr> <tr> <td>mitochondrial membrane(s)</td><td>E</td></tr> <tr> <td>production of subunits of ribosomes</td><td>nucleolus / nucleoli</td><td>C</td></tr> <tr> <td rowspan="4">active transport of ions</td><td>cell <u>surface</u> membrane ecf for cell membrane</td><td>A</td></tr> <tr> <td>or tonoplast / vacuolar membrane</td><td>F</td></tr> <tr> <td>or mitochondrial membrane(s)</td><td>E</td></tr> <tr> <td>or nuclear envelope A nuclear membrane(s)</td><td>B</td></tr> <tr> <td>aerobic respiration</td><td>mitochondrion / mitochondria</td><td>E</td></tr> </tbody> </table>	function	name of cell structure	letter on Fig.1.1	gas exchange	cell <u>surface</u> membrane I phospholipid bilayer	A	mitochondrial membrane(s)	E	production of subunits of ribosomes	nucleolus / nucleoli	C	active transport of ions	cell <u>surface</u> membrane ecf for cell membrane	A	or tonoplast / vacuolar membrane	F	or mitochondrial membrane(s)	E	or nuclear envelope A nuclear membrane(s)	B	aerobic respiration	mitochondrion / mitochondria	E	4
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1(b)(i)	endocytosis / pinocytosis ; R phagocytosis	1																							

Question	Answer	Marks
1(b)(ii)	any three from: 1 <u>hydrolysis</u> / use water to break bonds ; A ref. to hydrolytic enzymes 2 (action of) protease(s) / peptidases / proteolytic enzyme(s) ; 3 peptide bonds broken ; R if part of a list with other bonds not found in proteins, such as glycosidic, ester, phosphodiester 4 to form, peptides / amino acids ;	3
1(c)	1 formation of phagolysosomes / described ; A fusion with, phagosomes / phagocytic vesicle or vacuole 2 breakdown / digestion, of, pathogen / bacteria / microorganisms / AW ; A kill / destroy, pathogen / bacteria / microorganisms / AW one from: 3 (by) two named enzymes ; e.g. lysozyme, protease, lipase, carbohydrase, nuclease 4 any two named substrates that are hydrolysed ; e.g. polysaccharides, peptidoglycans, proteins, lipids, glycogen, nucleic acid(s) 5 to form, harmless / useful / soluble / AW, products ; I ref. to breakdown of organelles	2